Extensions of a nested ensemble Kalman filter

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Abstract

State-space models with a static parameter and a latent Markov process have an observed-data likelihood given by an integral over the latent path. Nested particle filters estimate the sequence of parameter posteriors, but a particle likelihood becomes expensive as the observation dimension grows. Replacing each likelihood increment by the Gaussian marginal of an ensemble Kalman forecast gives a nested filter for the posterior induced by that approximation. We compare an exact tree for the neighbour screen, a switch between a random walk and delayed acceptance that depends only on time, a nearest-neighbour surrogate used as the weight and a Rao-Blackwellised correction for a different linear Gaussian observation model. On an Ornstein–Uhlenbeck pilot the tree returns the same neighbours as a list. The switch has a lower cost per accepted move than either kernel alone. The surrogate weight uses fewer transition draws and has a larger error in the posterior means. The correction leaves the weight unchanged when the filter already uses the true observation model. A finite-difference Langevin proposal has a smaller effective sample size per transition draw than the random walk.

Keywords: delayed acceptance, Ornstein–Uhlenbeck process, Rao-Blackwellisation, sequential Monte Carlo, surrogate likelihood

1 Introduction

State-space models are used when a latent Markov process is observed with noise and the parameter θ is static. The likelihood of a series is an integral over the latent path, and that integral is rarely available in closed form. Sequential Monte Carlo squared (SMC2) propagates a cloud of parameter particles, reweights them by an unbiased particle-filter estimate of each likelihood increment, and moves them with a particle Metropolis–Hastings kernel when the effective sample size falls (Chopin et al. 2012). The variance of that estimate grows with the length of the series and, more sharply, with the observation dimension.

The ensemble Kalman filter replaces the forecast distribution by a Gaussian distribution whose mean and covariance are the sample moments of an ensemble, and it updates the ensemble with a Kalman shift. The likelihood contribution is the Gaussian marginal of that forecast. It is not unbiased for the true likelihood. Golightly et al. (2026) place this contribution inside the SMC2 recursion and move the parameter particles with one step of ensemble Markov chain Monte Carlo. The nested ensemble Kalman filter therefore targets the parameter posterior induced by the ensemble Kalman likelihood, denoted $\hat{\pi}_{\mathrm{enkf}}$, rather than the exact posterior.

The resample-move step dominates the cost. At time t the ensemble filter is rerun from the start of the series for every proposal. Golightly et al. (2026) screen proposals with a k nearest-neighbour surrogate of the ensemble Kalman log-likelihood. The same discussion points to a faster neighbour search, a choice among kernels, a gradient proposal and the use of the surrogate in the weight as well as in the screen. It also gives a Rao-Blackwellised weight for a nonlinear observation model. It also illustrates a regime of small parameter dimension and larger observation dimension. Those remarks are the starting point for the calculations below.

Our contribution is to name the target of each modification and to compare the modifications on a linear Gaussian pilot with a known likelihood. (i) The delayed-acceptance kernel is unchanged when a tree and a list return the same neighbours. (ii) A kernel chosen before the proposal, as a function of the time index alone, preserves $\hat{\pi}_{\mathrm{enkf}}$ when each candidate kernel does. (iii) A nearest-neighbour incremental weight targets the posterior under that surrogate, which we denote π_{knn}. (iv) When the true observation density equals the linear Gaussian density used by the filter, the Rao-Blackwellised weight equals the ensemble Kalman weight. A finite-difference approximation to the gradient of the ensemble Kalman log-likelihood can be used inside a Metropolis–Hastings proposal. On the pilot its effective sample size per transition draw is worse than that of the random walk, and we do not recommend it.

The pilot is a scalar Ornstein–Uhlenbeck diffusion. It uses 60 parameter particles, 16 observations and an ensemble of size 12, with four replicate series. These settings are smaller than the experiment in Section 5.1 of Golightly et al. (2026), and the gold-standard chains are shorter than the chains reported there.

The remainder of this paper is organised as follows. Section 2 describes the neighbouring algorithms and what each implies for the nested filter. Section 3 sets out the targets of the modifications. Section 4 reports the pilot, and Section 5 discusses what the pilot supports and what remains open.

2 Related work

SMC2 propagates a cloud of parameter particles, and each particle carries a particle filter. The incremental weight is an unbiased estimate of the likelihood contribution. Chopin et al. (2012, Proposition 1) write the extended target so that its θ-marginal is the exact parameter posterior for any fixed number of state particles. The rejuvenation kernel is particle Metropolis–Hastings on that extended target. Replacing the inner particle filter by an ensemble Kalman filter removes the unbiasedness on which this marginal property rests, and the marginal is then $\hat{\pi}_{\mathrm{enkf}}$.

Ensemble Markov chain Monte Carlo uses the same ensemble Kalman likelihood inside one Metropolis–Hastings chain on a fixed data set (Drovandi et al. 2022). One rejuvenation sweep of the nested filter is one iteration of that scheme, but the chain itself is not sequential. Even when the forecast is exactly Gaussian, the plug-in Gaussian density is a biased estimator of the idealised Gaussian predictive. An unbiased estimator of that Gaussian density does not restore the exact posterior of the state-space model.

A different sequential construction gives the parameter an artificial transition and resamples it with a shrinkage kernel, so that the filter does not collapse onto a single parameter value (Liu and West 2001). The particle ensemble Kalman filter of Katzfuss et al. (2018) weights parameter particles by an integrated ensemble Kalman likelihood. For a constant parameter they return to that kernel, because a point-mass proposal would otherwise leave fewer and fewer distinct particles. The nested filter keeps θ static and restores diversity by a Metropolis–Hastings move.

Delayed acceptance screens a proposal with a cheap approximation and then corrects the screen with the intended target (Christen and Fox 2005). If the parent kernel is irreducible for that target, the proposal is reversible, and the approximation is positive wherever the proposal is positive, the target remains invariant. Sherlock et al. (2016) store the expensive evaluations in a KD-tree and use an inverse-distance neighbour average as the screen, inside batch Markov chain Monte Carlo. Bon et al. (2021) put a surrogate inside the mutation of a sequential Monte Carlo sampler for a static model. In both cases the second stage evaluates the expensive target. Sherlock et al. (2016, Section 5.1) contrast this with algorithms that target the approximate posterior itself, and that is the distinction drawn for the surrogate weight in Section 3.

Katzfuss et al. (2018, Proposition 2) show that, in their Example 5, the variance of the log ensemble Kalman likelihood is of order n/N when the forecast members are drawn independently from the true forecast, with n the observation dimension in that example. Golightly et al. (2026)

cite this scaling. The argument is for that example and for independent forecast draws, and it does not cover the nested filter. Temfack and Wyse (2026) also replace the inner particle filter of SMC2 by an ensemble Kalman filter. Their observation variance depends on the state, they use an unbiased Gaussian density estimator, and they are explicit that the Gaussian assumption remains. The mutation is particle marginal Metropolis–Hastings. They do not use delayed acceptance or the Rao-Blackwellised weight studied here.

Adaptive sequential Monte Carlo can choose among kernels that each leave the current target invariant, with the tuning carried as an auxiliary variable and with kernels scored by expected squared jumping distance (Fearnhead and Taylor 2010). A later SMC2 construction switches between particle marginal Metropolis–Hastings and particle Gibbs (Botha et al. 2023). Particle Gibbs needs a conditional particle filter, which an ensemble Kalman filter does not supply, and the unbiasedness used in those arguments is not available for $\hat{\pi}_{\mathrm{enkf}}$. The switch in Section 3 is chosen before the proposal and depends on the series only through time.

Chen et al. (2021, Theorem 3.2) consider linear state transitions and a Gaussian initial distribution, and they prove that the gradient of the ensemble Kalman log-likelihood converges to the true score in L^p at rate $N^{-1/2}$. Their algorithm is gradient ascent and has no accept-reject step. A particle-filter gradient inside an uncorrected Langevin proposal (Rosato et al. 2024) is a derivative of a different likelihood. Section 4 reports a corrected finite-difference proposal on the Ornstein–Uhlenbeck pilot, where the exact score is available by other means, and gives the reason that proposal is set aside.

3 Methods

The observation model in the linear Gaussian case is

$$f_t^{(\theta)}(y_t \mid x_t) = \mathcal{N}(y_t; Hx_t, R). \tag{1}$$

An ensemble of size N is propagated by the state transition. The forecast mean $\hat{\mu}_{t|t-1}$ and covariance $\hat{\Sigma}_{t|t-1}$ are the sample moments, with divisor $N-1$. The ensemble Kalman likelihood contribution is

$$\hat{p}^{(\theta)}_{\mathrm{enkf},t}(y_t \mid y_{0:t-1}) = \mathcal{N}\big(y_t; H\hat{\mu}_{t|t-1},\, H\hat{\Sigma}_{t|t-1}H^\top + R\big). \tag{2}$$

This contribution is exact for the Gaussian forecast approximation and is not unbiased for the true predictive density. The product of these contributions, multiplied by the prior, defines $\hat{\pi}_{\mathrm{enkf},t}$ up to the ensemble randomness carried by the extended target (Golightly et al. 2026, Section 4.1).

3.1 Nested filter

The outer recursion follows Algorithm 6 of Golightly et al. (2026). At time t, each parameter particle is updated by one ensemble Kalman step and its weight is multiplied by (2). If the effective sample size is at least γM, the cloud is left as it is. Otherwise the particles are resampled with replacement, the weights are reset to $1/M$, and each particle is offered one Metropolis–Hastings proposal. The proposal is a random walk whose covariance is ζ times the sample covariance of the other resampled particles. The acceptance ratio uses the prior and the ensemble Kalman likelihood of the series so far, and the likelihood at a proposed value is obtained by rerunning the filter from the known initial condition. In the pilot, $\zeta^2 = 2.56^2/d_\theta$ and the ensemble size is held fixed.

The printed Algorithm 6 places the effective-sample-size test inside the particle loop. The surrounding text applies one resample-move step when the effective sample size of the updated cloud falls below γM. We follow the surrounding text: the weight update is completed for every particle, and the test is applied once.

3.2 Delayed acceptance and the neighbour search

The screen replaces both ensemble Kalman likelihoods in the acceptance ratio by an inverse-distance average of the k nearest stored log-likelihoods (Golightly et al. 2026, Section 4.2). If the screen accepts, a second ratio corrects back to the ensemble Kalman likelihood. The product of the two ratios is the plain Metropolis–Hastings ratio, so the kernel leaves the extended ensemble Kalman target invariant under the conditions of Christen and Fox (2005). The distance in Golightly et al. (2026) is described as some metric. The pilot uses Euclidean distance on $\log \theta$, after standardising each coordinate by the standard deviation of the stored cloud.

A linear scan and an exact KD-tree that return the same neighbours produce the same surrogate and therefore the same kernel. A different deterministic neighbour set changes the kernel. Provided the surrogate is positive wherever the proposal is positive, the second stage still leaves the ensemble Kalman target invariant, because that stage evaluates the ensemble Kalman likelihood. The pilot uses exact neighbours. At this size the comparison is the number of distances evaluated by the tree and by the list. The tree is rebuilt on every query, so the comparison is a count of distances rather than a wall-clock time.

3.3 A switch that depends only on time

While t is small the ensemble Kalman likelihood is a short filter, and a screen saves little. The pilot rule uses plain Metropolis–Hastings when $t \leq 8$ and delayed acceptance with $k = 3$ when $t > 8$, on a series of length 16. The choice is a function of t only and is made before the proposal. Each kernel preserves $\hat{\pi}_{\mathrm{enkf},t}$, and a mixture of kernels with the same invariant distribution preserves that distribution. The argument does not require the unbiased likelihood of Chopin et al. (2012), and it does not use the particle Gibbs kernel of Botha et al. (2023).

The criterion was fixed before any output was seen. It is the number of transition draws used inside rejuvenation, divided by the number of accepted proposals. The same two ingredients, the relative cost of the cheap approximation and the stage-two acceptance rate, appear in the efficiency calculations of Sherlock et al. (2021). Their high-dimensional diffusion limit is a separate calculation, and it is not used here.

3.4 Surrogate incremental weights

If the incremental weight is the nearest-neighbour prediction of (2), the weights target

$$\pi_{\mathrm{knn},t}(\theta) \propto \pi(\theta)\, \exp\bigl(\ell_{\mathrm{knn},t}(\theta)\bigr), \tag{3}$$

conditional on the stored cloud that defines $\ell_{\mathrm{knn},t}$. This is neither $\hat{\pi}_{\mathrm{enkf},t}$ nor the exact posterior. Equation (2) already differs from the true likelihood, and the surrogate differs again from (2).

In the pilot the first k particles form a reference set and are always updated with the ensemble Kalman filter. Every other particle takes the inverse-distance prediction of the reference increments and does not run a forecast for that weight. After resampling, the reference slots are re-evaluated, so that the screen is built from ensemble Kalman likelihoods. A proposal that passes the screen is still evaluated with the ensemble filter. No bound is available on the distance between $\pi_{\mathrm{knn},t}$ and $\hat{\pi}_{\mathrm{enkf},t}$. The diagnostic was also fixed in advance. At the terminal time it recomputes (2) for eight non-reference particles. The surrogate was to be called a poor approximation of the ensemble Kalman increment if the mean, over replicates, of the median absolute log difference exceeded 1.

Table 1: Ornstein–Uhlenbeck pilot. Kernel comparison, with means over four replicates. Cost per acceptance is the number of rejuvenation transition draws divided by the number of accepted proposals. ESS denotes the effective sample size.

Kernel	Abs. error	Draws	Cost per acceptance	ESS
Metropolis–Hastings	0.172	33855	310.4	39.6
Delayed acceptance	0.126	23970	221.6	43.3
Switch at $t = 8$	0.132	23457	214.7	36.7

3.5 Rao-Blackwellised weight

For a general observation density f, the filter can still be run under a linear Gaussian approximation g. The Rao-Blackwellised increment averages, over the analysis ensemble, the terms

$$\hat{g}^{(\theta)}_{\mathrm{enkf},t}(y_t)\,\frac{f^{(\theta)}_t(y_t \mid x^j_t)}{g^{(\theta)}_t(y_t \mid x^j_t)}. \tag{4}$$

The parameter target induced by these increments is denoted $\tilde{\pi}_{\mathrm{enkf},t}$. If $f = g$, each ratio equals 1 and (4) reduces to (2). The correction then leaves both the target and the estimate unchanged. The identity is for this linear Gaussian case. It does not by itself describe a nonlinear observation model.

The comparison fixed before the runs is the absolute error in the posterior means of $\log \theta_i$, per transition draw. The Rao-Blackwellised filter is preferred on that criterion when the ratio is smaller than the ratio for the plain filter run under the same g. When $f = g$, the identity shows that the correction cannot reduce the error.

4 Simulation studies

The latent process is the scalar Ornstein–Uhlenbeck diffusion in Section 5.1 of Golightly et al. (2026),

$$\mathrm{d}X_t = \theta_1(\theta_2 - X_t)\,\mathrm{d}t + \theta_3\,\mathrm{d}W_t,$$

with the Gaussian transition printed there. The truth is $\theta = (1, 2, 1)$, the initial condition is $x_0 = 10$, and observations at times $1, \ldots, 16$ follow (1) with $H = 1$ and $R = 0.1$. The prior is independent, with $\theta_1 \sim \mathrm{Gamma}(2, 2)$, $\theta_2 \sim \mathrm{Gamma}(5, 3)$ and $\theta_3 \sim \mathrm{Gamma}(2, 5)$, in the shape-rate parameterisation. The source paper writes $\mathrm{Gamma}(a, b)$ and does not name the parameterisation. All series below were simulated from this model, and all algorithms were coded in base R.

The nested filter uses $M = 60$, $N = 12$, $\gamma = 0.4$ and $k = 3$. The ensemble size does not adapt. Four replicate series are used. The gold standard for each series is a random-walk Metropolis–Hastings chain on $\log \theta$, targeting the exact Kalman likelihood, with 3000 iterations after a burn-in of 1000 and with proposal standard deviation 0.15. That chain is much shorter than the 10^6 iterations used by Golightly et al. (2026), and M, T and N are smaller than in their Section 5.1. The posterior means from the pilot should be read with that Monte Carlo error in mind.

The reported error is the absolute difference, averaged over the three coordinates of $\log \theta$, between the weighted filter mean and the gold-standard mean. Cost is the number of transition draws. Tables 1–3 report means over the four replicates.

On the pre-registered criterion the switch has the lowest cost per accepted move. Delayed acceptance is close to the switch and is cheaper per acceptance than the plain random walk, and its absolute error is the smallest of the three. The neighbour sets from the list and the tree agreed on every check, 491 checks per replicate on average. The list evaluated a distance to

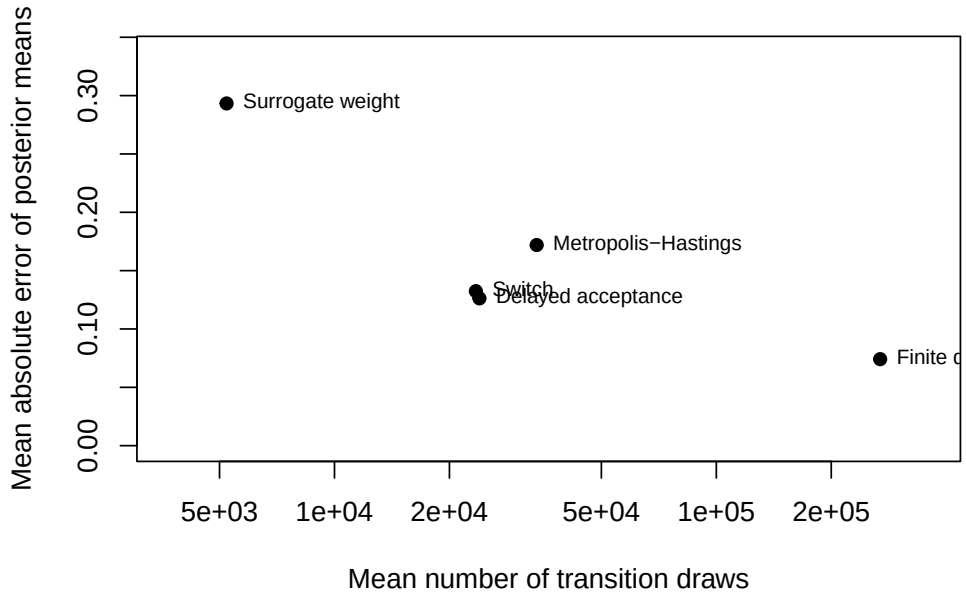


Figure 1: Ornstein–Uhlenbeck pilot. Mean absolute error in the posterior means of $\log \theta_i$ against the mean number of transition draws, over four replicates. The finite-difference proposal is included for comparison with the random walk.

Table 2: Ornstein–Uhlenbeck pilot. Sample variance of the log ensemble Kalman likelihood at the true parameter, from 20 independent ensembles.

d_y	N	Variance	Mean draws
1	8	14.060	128
1	16	2.879	256
5	8	24.935	640
5	16	10.661	1280

29460 stored points on average and the tree to 13422.5. Because the neighbour sets agree, the two searches define the same kernel on these runs. The tree evaluates fewer distances at this size. The count does not include the cost of building the tree.

Figure 1 places the runs of Table 1 on one scale, together with the surrogate-weight filter and the finite-difference proposal. The finite-difference proposal had mean absolute error 0.074 and mean transition draws 268602, against 0.172 and 33855 for the random walk. Effective sample size per transition draw was 0.000148, against 0.00123 for the random walk. The smaller error does not offset the extra draws under the pre-registered comparison, so the proposal is not taken forward. The acceptance probability included the Gaussian proposal density. The comparison is therefore one of cost for a kernel that still leaves $\hat{\pi}_{\mathrm{enkf}}$ invariant.

The surrogate-weight filter used 5217 transition draws on average. The mean, over replicates, of the median absolute log difference between the surrogate increment and a fresh ensemble Kalman increment was 0.277, below the threshold of 1. The mean absolute error was 0.293, which is larger than the error of the plain filter and of delayed acceptance. Most of the saving comes from particles that do not run a forecast. The target is (3), so the larger error is the discrepancy between that target and $\hat{\pi}_{\mathrm{enkf}}$.

Table 2 is the scaling comparison that the pilot can support. The five-dimensional series

Table 3: Ornstein–Uhlenbeck pilot. Rao-Blackwellised weights. The identity row uses $R_g = R = 0.1$. The other two rows use $R_g = 0.5$, whilst the data and the gold standard use $R = 0.1$. Error per draw is the ratio of the mean absolute error to the mean number of draws.

Filter	Abs. error	Draws	Error per draw
Plain, $R_g = 0.5$	0.369	22776	1.62×10^{-5}
Rao-Blackwellised, $R_g = 0.5$	0.169	36069	4.68×10^{-6}
Rao-Blackwellised, $R_g = 0.1$	0.121	31248	3.87×10^{-6}

consists of five independent copies of the same diffusion, observed with independent noise. Doubling N reduces the variance for both observation dimensions. The variance does not fall in proportion to the ratio of those dimensions. Proposition 2 of Katzfuss et al. (2018) concerns an example with independent draws from the true forecast. The ensemble used here is interacting, and the reported log-likelihood is a sum over the series. The four cells do not support a boundary at parameter dimension 10 or at observation dimension 10.

When $R_g = R$, the log ensemble Kalman increment and the log Rao-Blackwellised increment agreed on every replicate: the maximum absolute difference was 0. The correction does not change the weight in that case. When $R_g = 0.5$ and the true observation variance is 0.1, both densities are Gaussian and $f \neq g$. The Rao-Blackwellised filter then has the smaller error per draw, at a higher number of transition draws. The comparison uses two Gaussian variances. It is not a nonlinear observation model, and SMC2 was not run on the same series.

5 Discussion

The nested ensemble Kalman filter keeps an SMC2 recursion in problems where an unbiased particle likelihood is too expensive, at the cost of targeting $\hat{\pi}_{\text{enkf}}$ rather than the exact posterior. On the linear Gaussian pilot, delayed acceptance, the time switch and the Rao-Blackwellised identity behave as the arguments in Section 3 suggest, and the finite-difference proposal does not earn its cost.

Delayed acceptance, with the neighbour surrogate corrected by the ensemble Kalman likelihood, reduced the number of transition draws relative to a plain random walk and also reduced the absolute error, on this example. An exact tree returned the same neighbour sets as a list. It evaluated fewer distances at parameter dimension 3 and with 60 parameter particles. Whether that reduction remains once the tree is maintained, and whether it remains for a larger cloud, is open. The invariance argument above is for exact neighbours.

Choosing the plain kernel for the first half of the series, and delayed acceptance afterwards, lowered the cost per accepted move a little further than delayed acceptance alone. The rule is relatively ad hoc. Because it depends on the series only through time, the mixture of the two kernels preserves $\hat{\pi}_{\text{enkf}}$. A rule that looked at the proposed particle would sit inside the kernel, and no such rule was used.

The surrogate weight changes the target from $\hat{\pi}_{\text{enkf}}$ to π_{knn}. On the pilot the diagnostic stayed inside the threshold fixed in advance, the number of transition draws fell sharply, and the posterior means moved further from the gold standard than the means under $\hat{\pi}_{\text{enkf}}$. The surrogate weight is attractive only when that extra error is acceptable. A bound on the error is not available.

The Rao-Blackwellised correction leaves the weight unchanged when the observation model is already the linear Gaussian model inside the filter. The two log increments agreed exactly on the pilot. When the filter was run under a deliberately larger Gaussian variance, the correction reduced the error per draw. This is a narrow comparison between two Gaussian observation models. It does not establish the ordering, suggested by Golightly et al. (2026), of the plain nested filter, the Rao-Blackwellised filter and SMC2 under a nonlinear observation model.

The finite-difference proposal is the modification we drop. For linear transitions and a Gaussian initial law, a derivative of the ensemble Kalman log-likelihood converges to the true score as the ensemble grows (Chen et al. 2021, Theorem 3.2), and a Metropolis–Hastings correction can use a noisy finite difference without changing $\hat{\pi}_{\mathrm{enkf}}$. Forming the difference, and the reverse density, repeated the filter often enough that the effective sample size per transition draw fell below that of the random walk. An uncorrected Langevin proposal was not used.

The pilot has several limits of its own. The gold-standard chain is short, the parameter dimension is three, and the observation dimension in the main comparison is one. Table 2 is the only step towards the regime suggested by Golightly et al. (2026), and it does not confirm a threshold there. The Gamma priors were given a shape-rate reading that the source paper does not state. The series are simulated. The comparisons above already use a model with a known likelihood.

What remains open is a bound on the discrepancy between π_{knn} and $\hat{\pi}_{\mathrm{enkf}}$, a neighbour search whose wall-clock cost is measured at the cloud sizes used in applications and a nonlinear observation model on which the Rao-Blackwellised weight can be compared with SMC2.

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